

APT 3020 – Knowledge-Based Systems

Lab 2: Reasoning and Inferencing in Expert Systems

Explanation Report

1. Introduction

This report documents the reasoning and inferencing process of a rule-based expert system designed to advise students on academic outcomes. The system uses a knowledge base of facts and rules to produce conclusions such as scholarship eligibility, graduation status, academic probation, and dean's list candidacy.

2. Knowledge Base Overview

The knowledge base is stored in and contains two components:

Facts

Facts represent observable attributes of a student:

Fact Key	Description
gpa_above_3_5	GPA Above 3.5
gpa_between_3_and_3_49	GPA Between 3.0 and 3.49
gpa_below_3	GPA Below 3.0
attendance_above_80	Attendance Above 80%
attendance_below_80	Attendance Below 80%
no_disciplinary_cases	No Disciplinary Cases
has_disciplinary_cases	Has Disciplinary Cases
completed_prerequisites	Completed Prerequisite Courses
no_outstanding_fees	No Outstanding Fees
has_outstanding_fees	Has Outstanding Fees

Rules

Rules define the conditions that must be met for a conclusion to be drawn:

Rule	Conditions	Conclusion
Scholarship Rule	GPA > 3.5, Attendance > 80%, No Disciplinary Cases	Eligible for Scholarship
Graduation Rule	GPA > 3.5, Prerequisites Done, No Outstanding Fees	Eligible for Graduation
Graduation Rule (3.0–3.49)	GPA 3.0–3.49, Prerequisites Done, No Outstanding Fees	Eligible for Graduation
Probation Rule	GPA < 3.0	Academic Probation
Registration Block Rule	Has Outstanding Fees	Registration Blocked
Dean's List Rule	GPA > 3.5, Attendance > 80%	Dean's List Candidate

Reasoning Methods

3.1 Forward Chaining

Forward chaining is a data-driven approach. The system starts with the known facts about a student and checks each rule to see if all its conditions are satisfied. Every rule that matches fires and adds its conclusion to the result set.

Process:

1. Collect student attributes as a set of facts
2. Iterate through all rules in the knowledge base
3. For each rule, check if all conditions exist in the fact set
4. If yes, add the conclusion to the output
5. Repeat until all rules are evaluated

3.2 Backward Chaining (Bonus)

Backward chaining is a goal-driven approach. The system starts with a specific goal (e.g., "Eligible for Scholarship") and works backwards to verify whether the supporting conditions exist in the current fact set.

Process:

1. Define a list of possible goals
2. For each goal, find the rule that produces it
3. Check if all conditions of that rule are present in the student's facts
4. If yes, the goal is confirmed; otherwise it is not met

4. Explanation Facility

The explanation facility traces why each conclusion was reached by reporting the rule that was triggered and the specific facts that satisfied its conditions.

Format:

```
✓ [Conclusion]
Because ([Rule Name]):
- [Fact 1]
- [Fact 2]
- [Fact 3]
```

This allows the system to be transparent and auditable, showing not just what was concluded but why.

5. Test Cases and Explanations

Test Case 1 — Alice (High Achiever)

Input Facts:

- GPA: 3.8 → gpa_above_3_5
- Attendance: 90% → attendance_above_80
- No Disciplinary Cases → no_disciplinary_cases
- Completed Prerequisites → completed_prerequisites

- No Outstanding Fees → no_outstanding_fees

Rules Triggered:

Rule	Conditions Met	Conclusion
Scholarship Rule	GPA > 3.5 ✓, Attendance > 80% ✓, No Disciplinary ✓	Eligible for Scholarship
Graduation Rule	GPA > 3.5 ✓, Prerequisites ✓, No Fees ✓	Eligible for Graduation
Dean's List Rule	GPA > 3.5 ✓, Attendance > 80% ✓	Dean's List Candidate

Conclusions:

- Eligible for Scholarship
- Eligible for Graduation
- Dean's List Candidate

Test Case 2 — Bob (Average Student)

Input Facts:

- GPA: 3.1 → gpa_between_3_and_3_49
- Attendance: 75% → attendance_below_80
- No Disciplinary Cases → no_disciplinary_cases
- Completed Prerequisites → completed_prerequisites
- No Outstanding Fees → no_outstanding_fees

Rules Triggered:

Rule	Conditions Met	Conclusion
Scholarship Rule	GPA > 3.5 ✗	Not triggered
Graduation Rule (3.0–3.49)	GPA 3.0–3.49 ✓, Prerequisites ✓, No Fees ✓	Eligible for Graduation
Dean's List Rule	GPA > 3.5 ✗	Not triggered

Conclusions:

- Eligible for Graduation

Test Case 3 — Charlie (At-Risk Student)

Input Facts:

- GPA: 2.5 → gpa_below_3
- Attendance: 60% → attendance_below_80
- Has Disciplinary Cases → has_disciplinary_cases

- Prerequisites Not Completed
- Has Outstanding Fees $\rightarrow$ has_outstanding_fees

Rules Triggered:

Rule	Conditions Met	Conclusion
Probation Rule	GPA < 3.0 ✓	Academic Probation
Registration Block Rule	Outstanding Fees ✓	Registration Blocked

Conclusions:

- Academic Probation
- Registration Blocked

6. Comparison: Forward vs Backward Chaining

Aspect	Forward Chaining	Backward Chaining
Direction	Facts $\rightarrow$ Conclusions	Goal $\rightarrow$ Facts
Approach	Data-driven	Goal-driven
Use case	When all facts are known upfront	When checking a specific outcome
Conclusions	Finds all possible conclusions	Verifies one goal at a time
Efficiency	May evaluate unnecessary rules	Focused — only checks relevant rules

Both approaches produced identical conclusions for all three test cases, confirming the correctness of the implementation.

7. Conclusion

The expert system successfully demonstrates knowledge representation, rule-based reasoning, and inference using both forward and backward chaining. The explanation facility ensures transparency by tracing each conclusion back to the rules and facts that triggered it.